Module 2

- Python Rules
  - naming cannot begin with a number
- Control Flow → { if / elif / else
  while loop: 有條件的 repeat
  for loop = repeat 每一個.
  break / continue / pass
  - break = 馬上 exit.
  - continue = skip and move on the next iteration.
  - pass = do nothing in that block.

- Data Structure.
  - Ordered
    - changable — List = [ ]
    - un " — Tuple = ( )
  - Unordered
    - changeable — Set = { }
    - un " — Dictionary = { }

- 名詞釐清
  - method 方法 = object.method( )
    - * 屬於物件的特定用法.
    - * 有時會改變 object.
  - function 函數 = def / lambda
    - * 不改變 object

- List Methods (for Tuple Too)
  - append( ) 新增到此數後
  - count(x) x 出現 n 次
  - extend (x) x 可 = set
  - index (x) x 首次出現的位置.
  - insert (i, x) 在 i^{th} 位插入 x.
  - pop 移除首項.
  - remove (x) 移除 x.
  - reverse
  - sort(x) (reverse = True) → descend

- sort() : 改變的method | 改變原list | 不回傳
  sorted() : 全域 | 不改 | 回傳新list

- Set Methods

. add	. a.union(b)	a|b 聯集
. clear 清除all	. a.update(b)	a|=b 把b加入a
. remove	. a.intersection(b)	a&b 交集
. pop 移除任意元素	. a.intersection_update(b)	a&=b a更新成交集

- Dictionary methods

  . keys() → return key
  . values() → return value
  . items() → return both key and value

- iterable / map()

  iterable : 可迭代的sequence.
  map(function, iterable) : 把某個函數套用到iterable
  * 有回傳iterator (map物件) 要釋放記憶體

- Lambda

  lambda x : expression

NumPy

- attribute - 類似methods但描述物件的狀態.

. shape	. dtype	. flags
. ndim dimension數量	. T transpose	
. itemsize length in bytes	. real real part	
. size element數量	. imag imaginary part	

- unary functions
  - abs
  - sqrt **0.5
  - square **2
  - exp
  - log, log10, log2, log1p
  - sign
  - ceil 每個element舉離整數的最小整數
  - floor 〃 ↓
  - rint round to int
  - modf return { int / fractional } 兩個array
  - isnan boolean 確認是否為NaN
  - isfinite 〃 finite
  - logic_not = 邏輯上的反

- Binary Function
  - add
  - subtract
  - multiply
  - divide
  - power
  - maximum
  - minimum

- .where
  .where(cond, xarr, yarr)

PANDAS

- { Rows = individual observation ⟷
  Columns = variables ↕

- Series

- DataFrame
  - access to rows 取某几横列. → [中括号]
    ① loc[ ]
    用 label name 取
    (都用" "存取、可用 Boolean)
    ② iloc[ ]
    用数字位置取 (从0开始算)
    (不知名字但知道位置)

  - access to column
    _____.["column name"]
  - filter = Boolean
    *用取值 df[df["column"] condition]

- Manipulation
  - drop { axis=1 ↓ drop一直行.
    axis=0 → drop一横列.
  - rename (columns = {" " : " "})
  - sort_values (, ascending= T or F)
    ↳ 卡.sort() 卡 sorted()
  - apply (lambda x: _____)
    apply(def)
  - map (_____)

Module 3

① Ingest and integrate

- Load and Inspect Individual data sources.
- Correct Initial data errors.
- Join and merge data.
- Post-merge validation and documentation

② Shape and structure

- Standard structure (2 principle)
- Reshape
- Exceptions
- implementation

③ Validate and document

LOAD DATA

override data type

- pd.read_ file type
  * assume first row = column name
    head=none to treat first row as data.

JSON 格式

- 数据结构
  - object 对象 = key:value pair
    {大括号}，不同 pair 用逗号.
    (有 key:value 的部分)
  - array 数组 = multiple value array (in order)
    [中括号] ⇒ 表的表示法
    (有重复/生的 data)

Preparation STEP1 检查数据的完整度.

① Inspect Structure ↳ Pandas Function:

- .shape
- .columns / .dtypes
- .info()

② Preview Sample rows

- .head()
- .sample(n) = random rows
- .tail() = 同学们都数据的 info.

③ Check for missing values

- .isnull().sum() = Total missing value
- .notnull().mean() = proportion of non-missing value.

④ Assess duplicates

- .duplicated.sum()
- df[df.duplicated(subset = key columns)]
  = check duplicates on key

※print("字{查入的df名}")
↳ 大括号.

⑤ Spot common formatting or type issues

Preparation STEP 2 決定哪些欄位要被修正.

- to_numeric(_____, errors=_____, downcast = _____)
  ↳縮小資料型式
  ↓
  raise = 遇到錯誤就報錯.
  coerce = 錯誤 = NaN.
  ignore = 有錯誤就全不修改.
- pd.to_datetime.
- reorder column ⇒ q2 = q2[q1.columns]
- drop dummy header ⇒ q2 = q2[q2["column name"] != "Content"]
- .rename(columns={"原名":"新名"}, inplace = ___)
  ↳ True 直接修改原物件
  False 回傳新的

- .str.strip() / .str.upper()
- .drop_duplicates

Preparation STEP 3 不同來源的dataset整理，連結keys.

- Vertical：直向的加入same column的更多 data
  = pd.concat( ), axis = 0.
  - axis = 0 向上下擴 / add more row
  - axis = 1 向左右擴 / add more column
- Horizontal：橫向加入更多屬性的 column
  = pd.merge(被加入者, 加入者, ...)
  - Join
    - Inner：
    - Outer：
    - left：
    - Right：
  - on = " "
  - left_on = __, right_on = ____
    (不同table名稱不同時)
  - how = inner ...(連結方法)

Preparation STEP 4 確認前面的modify沒問題

- check row counts：比較merge前後的行列數字
- check for missing values：檢查出NaN（先做identify即可）
- check for duplicated keys：

Module 3 – shape and structure

- Principle
  - Structural – unit of analysis = unit of observation
  - Independence – each row is independent.

Unit of Analysis = primary object that answer the question "what are we analysing?"
對象

Unit of Observation: the object which data is collected.

- Reshape: 使unit of analysis = observation 對應一致

- Difficulties
  - Repeated observation over time
    – 對象, 多個 instance
  - Multiple observation (due to hierarchical)
    – 對象, 多個 observation

- Exceptions ⇒ could ignore the principle!
  need to take into consideration during analysis.

- wide table ≒ spread data (standard format)
- tall table ≒ Tidy data

Tidy
- exploration and modeling easier.
- expose patterns of dependency

types
- time/sequence
- hierarchical
- spatial/relational

- tall table ⇒ wide table
  - df.pivot(.....) 表变宽 (分類值变新欄位)
    ↳ 看矩陣化結構
    - index = 指定成索引
    - columns: 指定哪欄的值 成為新的欄位名
    - values = 指定放入值的 value 是哪一欄
  - reset_index() 避免任何一個 column 变成 index.
- wide table ⇒ tall table
  - .groupby(.....)
  - .melt(.....) 宽变長 (多欄压縮成兩欄)
    ↳ 做統計/繪圖用.
    - id_vars = 保留不变的.
    - value_vars = 压扁成变长的.
    - var_name = 压扁的新欄位名稱
    - value_name = 值的 column.
  - split
  - reorder

EDA

① variable classification

- Qualitatives / categories
  - Nominal 有分類、無優劣、單純分別用
  - Ordinal 有分類、有順序關係
- ↑ 平均有意義 meaning ↓
- Quantitatives / measures
  - Interval (區間) 差距有意義、zero 相對
  - Ratio (比例) 〃 、TRUE ZERO
  - → discrete / continuous

② Univariate analysis

- 一組資料的 two main features
  - ① Location = data 的集中點在哪
    mean, Trimmed and Winsorized mean, median, mode
    Geometric mean harmonic mean
  - ② Variability = 數據分散、聚合的程度
    Range, Variance, SD, Coefficient, IQR.
- Skewness — asymmetry
- kurtosis — Tailedness
- visualization
  histogram / Boxplot / Violin Plot
  ↓

* kernel density estimate
  ↳ ≈ probability density function

* Bin size selection
  { square root rule: Bins ≈ $\sqrt{\text{numbers of observation}}$
  Sturges' rule: Bins = $1 + \log_2(\quad n \quad)$

* ensure y-axis starts at zero.

- categorization
  Barchat / Pie chart.
  * high cardinality category 高基数类别.
    ↳ combine less frequent into "other"

- Diagnostic
  ① missing value
  - identify the cause of missingness.
  - if there are patterns "informative missingness"

how to deal { Retain observation
Remove
Perform imputation (replace with estimated)

- Pattern
  Missing complete at Random (MCAR)
  完全無關的缺失.
  Missing at random (MAR) = conditional missing
  缺失與已觀測到的變數有關.
  在處理控制後 imputation.
  Missing not at random (MNAR)
  值本身就是缺失原因.

↓ (importance)

Determine

① MCAR? → { H_0 = MCAR, $H_1 \neq$ MCAR } chi-square test

② MCAR → Yes
→ No → MAR?
→ MNAR?
) could exist both.

② Outliers

detection [Z-score → $\bar{X} \pm 4 \sim 5\sigma$.
IQR → [25th percentile − 1.5IQR
75th percentile + 1.5IQR.

③ Bivariate

- use chi-square / correlation to prioritize

- measure / measure = two numeric variable
  correlation matrix, scatterplot
- measure / category = 比较多类别的 numeric
  Boxplot, Violin plot, Barchart
- category / category = 两类别的
  chi-square, standard residuals

④ multivariate

- diagnose multicollinearity / confounding.
- faceted visualization, enhanced scatter plot, color-coding bubble plots, 3D scatter plot, stacked bar charts.
- aggregated visualisation
  heatmaps, Treemaps, Mosaic plots
- high dimensional
  parallel coordinates plot, radar chart.

- diagnostic

① outliers

- mahalanobis distance
- high leverage observation
  outliers ⇄ HCO (不一定 / 不一定)
- Residual analysis
- Hat matrix
- cluster analysis

② multicollinearity

- compute VIF

$$VIF(\beta) = \frac{1}{1 - R^2_{X_j | X_{-j}}}$$

R^2 ↑ deno ↓ VIF ↑

VIF 越大越糟糕。

VIF 大于 5-10 ⇒ problematic

③ interaction effect / confounding.

- interaction effect

  兩變數可能非獨立,

  呈現變數間的關係→要分化.

- confounding.

  external variable 左幹擾 X, Y

  造成實驗的誤判結果→要處理.

Feature Engineering.

- single variable

[1] Adjust distribution
→ log / √ / Box-Cox.
(以后学随分布更好?)

[2] Make variable comparable across different units.
→ Z-score / Min-max normalization
(标准化的概念?)

[3] convert categories into numeric form for models
→ dummy encoding / one-hot / high cardinality.

[4]

[5]

- Multiple variable features (combine 2+ → new feature)
- Multiple row features (use info across observations)

Module 4

	Sample Statistic	Population Parameter
Mean	$\bar{x}$	μ
Standard deviation	s	σ
Proportion	p	π

Std error for $\bar{X}$

Sample mean formula: $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$

$Var(\bar{X}) = Var(\frac{1}{n}\sum_{i=1}^{n} X_i) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{n}{n^2}{\sigma_X}^2 = \frac{{\sigma_X}^2}{n}$

Reminder: $Var(aX) = a^2 Var(x)$

$SE(\bar{X}) = \sigma_{\bar{x}} = {}^{\sigma}/_{\sqrt{n}} = \sqrt{Var(\bar{X})}$

Increasing the size of the sample (n) causes a reduction of the standard error by a factor of $\sqrt{n}$

Std

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Wait! Why is this $(n-1)$ and not n like in the variance formula?

CI = point estimate ± margin of error

- CI for mean.

$$CI = \bar{X} \pm z^* \left(\frac{\sigma}{\sqrt{n}}\right)$$

$$\downarrow$$

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

- σ未知時.

$$CI = \bar{X} \pm t_{n-1}(1-\alpha)\left(\frac{s}{\sqrt{n}}\right)$$

- proportion.

$$CI = p \pm z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}$$

Module 4

Hypothesis Test

- $\begin{cases} H_0 : \\ H_1 \end{cases}$ ← 主要目的皆為看有沒有足夠証據 Reject H_0.

ex= 想看是否 significant? (希望是)

$H_0 = \mu_0 = \mu$. 才可以 Reject.

- Four Possible Outcome

Jury Decision	Reality: H_0 true: defendant did not commit crime	Reality: H_0 false: defendant did commit crime
Reject H_0: find defendant guilty	Type I Error ("false positive")	Correct decision
Do not reject H_0: find defendant not guilty	Correct decisions	Type II error ("false negative")

One Sample

$$\begin{cases} H_0 : \mu \geq \mu_0 \\ H_1 : \mu \leq \mu_0 \end{cases}$$

T-test ⇒ Calculate mean

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} = \frac{\text{observed} - \text{Hypo}}{\text{Standard error}}$$

- $\begin{cases} S>9 & t\text{-test 檢查條款.} \\ S>1 & t\text{-test} \end{cases}$ mean.

① 看题目问什麼？

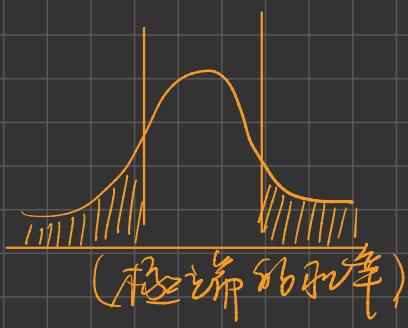
We want to test ___Ha___?

是否和变小 → $\mu < \mu_0$ left tailed.

" 变大 → $\mu > \mu_0$ right tailed.

② 算 t-test

③ 查 p-value → p-value 定義：



H_0 為真情況下，

有產生更極端值的机率？

(left tailed 只有是測 $\mu < \mu_0$)

④ 和 Bar(α) 做比較。

(以 left tailed 舉例)

H_0 沒有變小

如 $\alpha = 0.05$

$p = 0.08$

$p > \alpha$ 發生机率不高 → Cannot Reject

$p \leq \alpha$ 太低 → Reject

Z-Test ⇒ Calculate proportion

$$Z = \frac{P - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}} = \frac{\text{sample} - \text{hypo}}{\text{Sample 的 standard error}}$$

$$P = \frac{\text{number of success}}{n}$$

- two tailed：想看双边的极端值
  - ↳ 沒有指明的话，一律使用 Two tailed（保险）
- one tailed：想看单边的特定极端值。（另一边无意义）

Two Sample

$$\begin{cases} H_0 = \mu_1 \gtreqless \mu_2 \\ H_a = \mu_1 \lessgtr \mu_2 \end{cases}$$

T-test

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} - \frac{S_2^2}{n_2}}} \qquad \text{oldof} = (n_1 - 1) \,\&\, (n_2 - 1)$$

Paired T-test

→ 看同一對象在兩种狀態下，平均差是否相同？

$$\begin{cases} H_0 = \mu_D = 0 \\ H_a = \mu_D \neq 0 \end{cases} \qquad T = \frac{\bar{D}}{S/\sqrt{n}}$$

Z-test

$$Z = \frac{P_1 - P_2}{\sqrt{P_{pooled} \cdot (1 - P_{pooled}) \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \qquad P_{pooled} = \frac{X_1 + X_2}{n_1 + n_2}$$

Multiple Sample

$$\begin{cases} H_0 = \mu_1 = \mu_2 = \mu_3 = \cdots \\ H_a = \mu_1 \quad \mu_2 \quad \mu_3 \text{ 其中一個不符合條件} \end{cases}$$

ANOVA ⇒ Compare means across groups.

$$F = \frac{MSG}{MSE} = \frac{\text{mean square between group}}{\text{mean square error}}$$

$$= \frac{\text{Between group variance}}{\text{Within sample variance (Error Variance)}}$$

$$MSG = \frac{\sum_{i=1}^{k} n_i (\bar{X}_i - \bar{X})^2}{k-1} \leftarrow \text{ddof (K group)}$$

$$MSE = \frac{\sum_{i=1}^{k} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2}{n_i - k} \leftarrow \text{ddof}$$

$\Rightarrow$ Sample 的加權平均

One Way ANOVA : Single Variable 的比較組間平均

	Df	Sum Sq	Mean Sq	F value
Group	$df_G = k-1$	$SSG = \sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2$	$MSG = \frac{1}{df_G} SSG$	$F = \frac{MSG}{MSE}$
Residuals	$df_E = n-k$	$SSE = \sum_{i=1}^{k} (n_i - 1)s_i^2$	$MSE = \frac{1}{df_E} SSE$	
Total	$df_T = n-1$	$SST = \sum_{i=1}^{n} (x_i - \bar{x})^2$		

Table 1: Formulas for ANOVA calculations.

① One-Way ANOVA table (最經典的那種)

Source	SS	df	MS	F	p
Between Groups	SS_B	k−1	MS_B	F	p
Within Groups	SS_W	N−k	MS_W		
Total	SS_T	N−1			

Two way ANOVA：兩個variable的 (A, B)，有交互作用.

② Two-Way ANOVA table (多了 A, B, 交互作用)

Source	SS	df	MS	F	p
Factor A	SS_A	a−1	MS_A	F_A	p_A
Factor B	SS_B	b−1	MS_B	F_B	p_B
A×B	SS_{A×B}	(a−1)(b−1)	MS_{A×B}	F_{A×B}	p
Error	SS_E	N−ab	MS_E		
Total	SS_T	N−1			

Chi-square ⇒ frequency in different category.

$$\chi^2 = \sum_{i=1}^{k} z_i^2 = \frac{(\text{Observed} - \text{Expected Frequency})^2}{E}$$

$$= \frac{(O-E)^2}{E}$$

$$= \sum\sum R^2$$

Pearson Residual.
計算各cate用的頻率.

$$R = \frac{(O-E)}{\sqrt{E}}$$

Chi-Square Test Type	Purpose	Example
Goodness-of-fit test	Compares observed vs. expected frequencies for one categorical variable	Checking if customer preference for product colors matches expected proportions
Test for independence	Checks if two categorical variables are associated	Does education level influence voting behavior?
Test for homogeneity — 致性	Compares distributions of a categorical variable across multiple groups	Do regions have the same preference for mobile phone brands?

observed − expected.
X_1, X_2的關聯性.
不同因子下的分佈一致性.

Resampling.

Bootstrap.：有放回sample取量.

用以估计CI. SD. SE.

Permutation Test：取代假设检定.

与 Two Sample.

看A.B tag有没有significant.

mask掉A.B.

随机分给A.B.

看distribution分得如何.

Variable Type	Hypo Test	Reason
Continuous	two sample t-test	mean differ
flag	two sample z-test	proportion differ
multi-nomial	χ^2 test?	homogenity of proportion

判断 logic

- mean
  - single t
  - two t / paired-t
  - multi ANOVA.
- proportion
  - single z
  - two z
  - multi chi-square.

Module 5 SQL

query order

- SELECT ⑥
- FROM ①
- JOIN ②
- WHERE ③
- GROUP BY ④
- HAVING ⑤
- ORDER BY ⑦
- LIMIT / OFFSET ⑧

operation order

- FROM
- JOIN
- WHERE
- GROUP BY
- HAVING
- SELECT
- ORDER BY
- LIMIT

- FILTER — WHERE
  - Equality
  - Inequality:
    - Range = WHERE ___ BETWEEN ___
    - Membership = WHERE ___ IN ___ or Subqueries.
    - Matching = WHERE ___ LIKE ___
      - partial string matches — / %.
      - NULL = WHERE ___ IS NULL

- CASE

```
CASE
    WHEN
        THEN
    ELSE
END AS
```

JOIN

① SELECT
FROM
JOIN __ ON ____
JOIN __ ON ____

② SELECT
FROM
JOIN
(SELECT
⋮
)
ON

JOIN : combine columns. (水平方向)

UNION : combine rows (垂直上下)

UNION : compound 原本獨立的 query.

- UNION
- UNION ALL (有 duplicate row)
- INTERECT
- EXCEPT

GROUPING : GROUP BY

* GROUP BY 後的 select 只能選原表格中的 attribute /
max( ), min( ), count( ), sum( ), average( )

- Date and time

CURRENT DATE
- CURRENT_DATE()
- CURRENT_TIME()
- CURRENT_TIMESTAMP()
- YEAR(), month(), ...

DATE FORMATTING
- TO_DATE(string, format)
- DATE_TRUNC() 切到想要的粒度.

DATE INTERVALS (P.206)
- DATEDIFF
- DATEADD

Subquery

① return single row / single column.
→ use in equality / inequality / comparison.
= != >, >=

② return single row / multi column
→ use in IN / NOT IN

③ return multi columns.
→ 列出 steps 一個個拼湊.

- Correlated subquery.
子查詢有依賴外界的資訊, 無法獨立執行.
常見於每一列逐列比較的狀況.

CTE

```
WITH name (column1, .....) AS
(
    subquery ...
)
-- main query.
SELECT
    ⋮
```

使用時機：

① query有多個運算步驟

② 結果須使用>=2次

③ 精簡query.

④ 多層GROUP BY.

WINDOW FUNCTION

- 複雜版的 Grouping.

```
SELECT ____ [ ] ← 插入在這裡
FROM
    ⋮
```

WINDOW FUNCTION OVER(

PARTITION BY ____

ORDER BY ____ ____

FRAME

) AS name

- RANK()
- ROW_NUMBER()
- SUM()
- AVG()
- LAG()
- LEAD()
- ⋮

{RANGE | ROWS} BETWEEN ____ AND ____

{ n / UNBOUNDED + PRECEEDING / FOLLOWING
  current row

WINDOW FUNCTION

- RANKING
  - ROW_NUMBER() 1, 2, 3, 4, 5, 6, ...
  - RANK() 1, 2, 3, 4, 4, 4, 7, 8, 9
  - DENSE_RANK() 1, 2, 3, 4, 4, 4, 5,
  - PERCENT_RANK()
- extra: NTILE()
- VALUE
  - LAG (往前看)
  - LEAD(column, offset, default) (往後看)
  - FIRST_VALUE()
  - NTH_VALUE()
- AGGREGATE
  - AVG()
  - MAX()
  - MIN()
  - SUM()
  - COUNT()